

Chapter 147

Channel Splitting and Image Compositing

This chapter begins by showing node structures you can use to isolate, split apart, convert, and recombine the image channels of clip in different ways. This includes ways of using the additional image channels that are provided in some types of media.

The second part of the chapter shows ways that you can do image compositing right in the Color page, with techniques for using external mattes for compositing, and using qualifier keys to do green- and blue-screen compositing against other layers in the Timeline.

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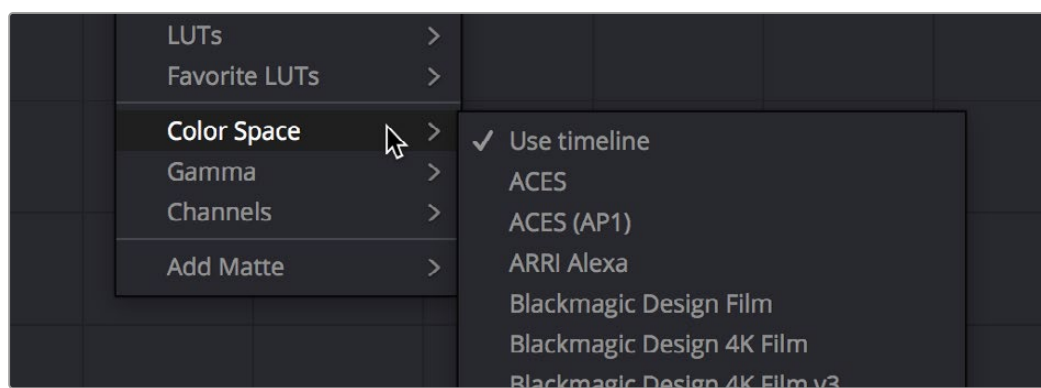
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Isolating, Splitting, and Converting Color Channels

DaVinci Resolve provides two different methods of making channel-specific adjustments, depending on whether you need to apply an adjustment to just one channel within a single node, or apply separate adjustments to all three channels across several nodes.

Enabling, Disabling, and Converting Node Channels

Within the contextual menu of each node in the Node Editor is a series of four options:



Expanded choices for choosing a color space and gamma for image processing within a node, and for disabling channels

While the ability to change the color space in which a particular node's operations work from the RGB default has been available for many versions, the list of available color spaces was greatly expanded in DaVinci Resolve 15 (all the previous options such as Lab (CIE), HSL, and YUV are still there). Additionally, you have the option of choosing the gamma that node works with as well, with a similarly long list of options.

Choosing a node-specific color space and gamma does not directly alter the image, as with the Color Space Transform ResolveFX plugin. Instead, changing a node's color space and gamma alters what type of image channels the red, green, and blue controls affect, and how the adjustments you make within that node are applied. For example, this lets you make a temperature adjustment with a node's gamma set to Linear, which in some instances may be mathematically advantageous.

Additionally, three checked Enable Channel 1–3 options let you turn individual channels off or on, limiting which channel that node's adjustments will actually affect.

In the following example, you'll see how to use these features to selectively sharpen just the Y' (luma) of an image without affecting the chroma, which can be a more subtle effect than simply sharpening the entire image.

To use channel disabling and color space conversion to sharpen luma only:

- 1 Add a node with which to apply the sharpening you want to the current clip.
- 2 Right-click the new node, and choose YUV from the Color Space submenu of the contextual menu.
- 3 Apply sharpening by doing one of the following:
 - Using that node's contextual menu, uncheck Enable Channel 2 and Enable Channel 3, which correspond to the U (Cb) and V (Cr) channels, leaving only Channel 1 (Y) enabled. Then, open the Blur palette, and drag the ganged Radius sliders down to sharpen the Y channel.
 - You can also just open the Blur palette, ungang the Radius sliders, and drag the red slider down to sharpen the Y channel, since any control with three gangable sliders will automatically assign those sliders to whichever channels are used by the currently selected Color Space.



(Before) The original image, (After) Sharpening applied to only the Y' channel of the image

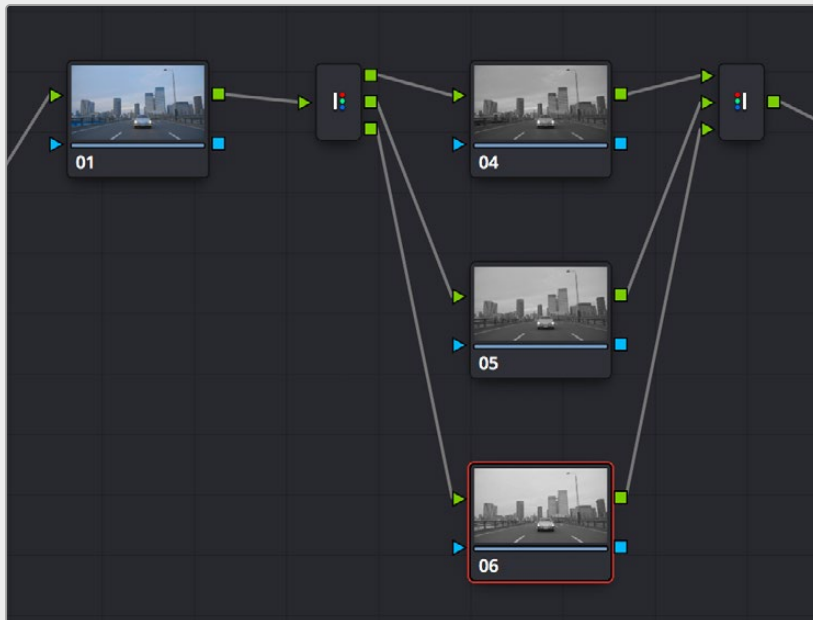
As you can see, while the Blur palette ordinarily provides separate R, G, and B controls that can be ungangd from one another, the Color Space submenu lets you apply sharpening to the channel definitions of other colorspace, providing many other corrective and creative possibilities with the same controls.

Splitting Channels with the Splitter/Combiner Nodes

Another method of applying corrections to individual color channels is using the Splitter/Combiner nodes, which break the Red, Green, and Blue channels apart into separate node tree branches, each capable of accepting multiple serial and parallel nodes of image adjustment. There are two ways you can create a Splitter/Combiner node structure.

Methods of adding Splitter/Combiner nodes:

- **To add a premade Splitter/Combiner node structure:** Choose Color > Nodes > Add Splitter/Combiner Nodes (Option-Y). A Splitter and Combiner node appear already connected to three Corrector nodes, one for each color channel.
- **To manually construct a Splitter/Combiner node structure:** Right-click in any empty area of the Node Editor, and choose from the Add Node submenu to create a Splitter node, a Combiner node, and three Corrector nodes, wiring them together as necessary to create the desired effect.



The Splitter Combiner node structure

The Splitter node takes an incoming image and provides individual outputs for each color channel (top/red, middle/green, bottom/blue). When you connect each of these outputs to a Corrector node, that color channel automatically connects to all three internal channels of that node, so that the default three nodes in a Splitter/Combiner structure are internally processing red/red/red, green/green/green, blue/blue/blue. The Combiner then pulls the Red, Green, and Blue channels out of each node connected to its three inputs, and reassembles them into a single RGB image again. Of course, what happens in between the Splitter and Combiner is entirely up to you.

One of the simplest ways of showing the possibilities of this node structure is to combine it with Node Sizing in the Sizing palette. Using Node Sizing, you can correct for misaligned channels in older archived video, or create channel misalignments for creative looks. In the following example, a simple Node Sizing adjustment to a split Blue channel will create an interesting “prism vignette” effect.

To use Node Sizing on individually split channels:

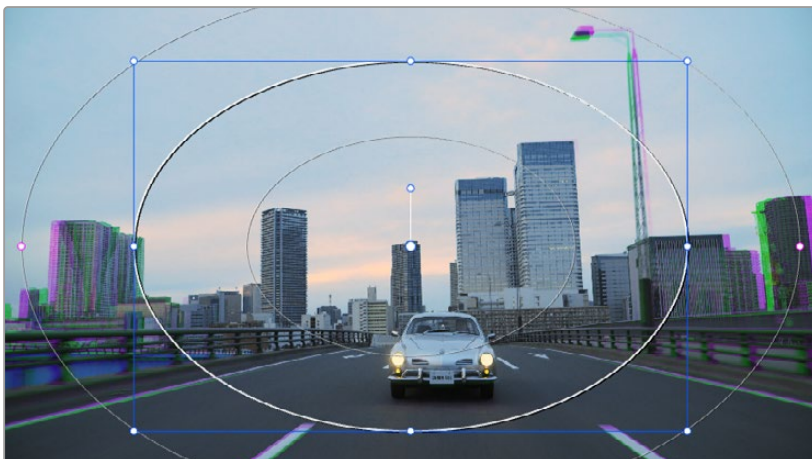
- 1 Choose Nodes > Add Splitter/Combiner Node (or press Option-Y) to add a premade Splitter/Combiner node structure to the Node Editor.
- 2 Select the middle of the three Corrector nodes appearing in between the Splitter and Combiner nodes (Node 5 in the following screenshot).
- 3 Open the Sizing palette, choose Node Sizing from the Mode drop-down, and then raise the zoom parameter by a small amount; this example uses 1.014.



Zooming just one color channel using the Splitter/Combiner nodes

As you can see in the screenshot, just the Green channel has been zoomed in on across the entire image. This is only possible using Node Sizing. This is cool, but not quite what you need.

- 4 Open the Window palette, add a Circular window, turn on its Invert control, and transform it to be a very soft, horizontally aligned vignette. This limits the individual scaling done to the Green channel to just the outer edge of the image, similar to a lens with excessive chromatic aberration.



The resulting channel split effect

You could easily keep going, applying varying amounts of zoom with different window shapes to each of the other color channels, and perhaps adding some blur to enhance the effect, but this example should show the creative potential available when using this technique.

Multi-Channel RED HDRx Support

The RED EPIC, SCARLET, DRAGON, and WEAPON cameras are all capable of shooting in an HDR mode that effectively “brackets” two different exposures of each frame. The resulting image data stores two channels of image data: the regular exposure, and a “highlight exposure” that’s underexposed by a user-selectable margin (+3, +4, +5, or +6 *f*-stops difference).

By default, the Input bar in the node graph feeds the regular exposure to your grade’s node tree. To take advantage of the additional “highlight” exposure, you can add an additional Source input to the node graph that feeds a second stream of image data that you can mix with the regular exposure in different ways.

NOTE: You can also take advantage of the highlight exposure of RED HDRx media by enabling and adjusting the Magic Motion controls in the Camera Raw palette.

To set up a node tree combining the normal and highlight HDR versions:

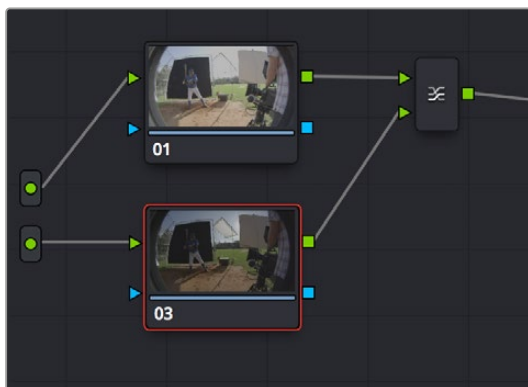
- 1 Using the first default node (Node 1), grade the image to see if you even need to use the alternate exposure that’s available. We’ll assume for this example that you do. In this example, the sky of the default exposure is pretty blown out, but there may be detail that can be retrieved using the highlight exposure.



Using HDRx images with the Layer Mixer

- 2 Create a Layer Mixer node by choosing Nodes > Add Layer (Option-L).

Two nodes are created, a Layer Mixer node that’s added after Node 1, and a third node (Node 3) that’s connected to a second RGB input of the Layer Mixer node in parallel to Node 1.



The Source currently supplies two input, but you can add a second source, which is the short exposure in the HDR image

- 3 Right-click anywhere within the node graph (except on a node) and choose Add Source from the contextual menu.

A second Source input appears underneath the first, which outputs the highlight exposure of the image as a separate image stream.



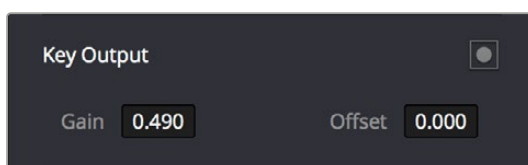
A node tree that uses a Layer Mixer node to mix two Corrector nodes, each connected to separate outputs for the regular and highlight exposures

- 4 Delete the connection between the top Source input and Node 3, and then connect the bottom Source input to the RGB input of Node 3.

Immediately, the image in the node thumbnail and Viewer updates to show the darker, underexposed version of the HDR image. This is because, by default, the Layer Mixer is mixing 100% of Node 3 over Node 1.

- 5 Select Node 3 and use one of the following two procedures to create a useful combination of the two exposures:

To create a mix of the two exposures, open the Key tab, and drag the Key Output Gain slider to lower the contribution of Node 3 to the overall image. Using dynamics (keyframing), this is a good way to animate a dissolve from the regular exposure to the highlight exposure if you're going from a dark environment to a bright environment in the same take and you want to create a smooth transition between both exposures.



Dissolving between the regular and highlight exposures of RED HDRx media using the Key Output Gain slider of a Corrector node connected to a Layer Mixer node

TIP: You can also use the Blend Type and Blend Bias parameters in the Camera Raw palette to blend the two exposures without building a dedicated node tree.

To use the highlight exposure to selectively put detail back into the image (for example, to retrieve blown-out windows), use a Power Window, HSL Qualification, or a combination of the two to isolate the region you want to retrieve in Node 3. (Be careful if you're using HSL Qualification to combine both exposures, as keyed edges can be tricky to blend.)



Isolating a region of the image to replace using the highlight exposure image of a RED HDRx clip



Final node display with isolations

A Note About RED HDRx Media and Performance

Since RED HDRx media records two complete streams of image data, DaVinci Resolve must simultaneously decode two separate tracks whenever you add a second Source input. (If you don't add the second Source input, only the first stream is decoded.)

Because of this, whether you're relying on your computer's CPU or GPU to decode RED media, you'll get half the performance when using the highlight stream of an HDR clip.

To improve performance, you can enable the Smart Cache by choosing Playback > Render Cache > Smart. This automatically caches all raw source clips to the current codec set in the Render Cache Format parameter of the Master Settings panel of the Project Settings. Alternatively, you can enable the User Cache, and cache all HDRx clips manually by right-clicking them and choosing Render Cache Fusion Output > On. DaVinci Resolve renders all cached clips while you're paused, so the next time you play a cached clip, it will play at full speed.

If you regularly use RED HDRx media, multiple high-end GPU cards are recommended for optimal decoding performance.

Introduction to Compositing Using the Alpha Output

While DaVinci Resolve can use the alpha channel of imported media for compositing, the Alpha Output in the Node Editor is an optional output that you can turn on to create clip transparency using operations inside DaVinci Resolve to create composites against other clips on lower video tracks. There are many ways of using the Alpha Output, but this section will focus on three examples.

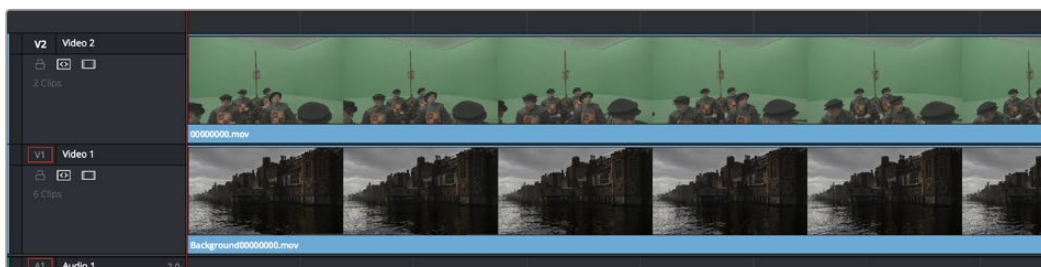
Using a Qualifier Key to Create Transparency

In this first example, the Alpha Output will be used to composite a green screen clip with a background plate.



A green screen clip, ready for compositing

In preparation for this composite, the background plate is on track V1 of the Timeline, and the green screen clip has been superimposed on track V2.



A superimposed green screen clip on track V2 of the Timeline, above a background plate on track V1, ready for compositing

To create a chroma-key composite in the Color page:

- 1 Right-click anywhere within the gray area of the Node Editor and choose Add Alpha Output to add the Alpha Output at the right of the node graph.

An Alpha Output appears underneath the Node Tree output at the right of the Node Editor.



The image Node Tree output with Alpha Output below

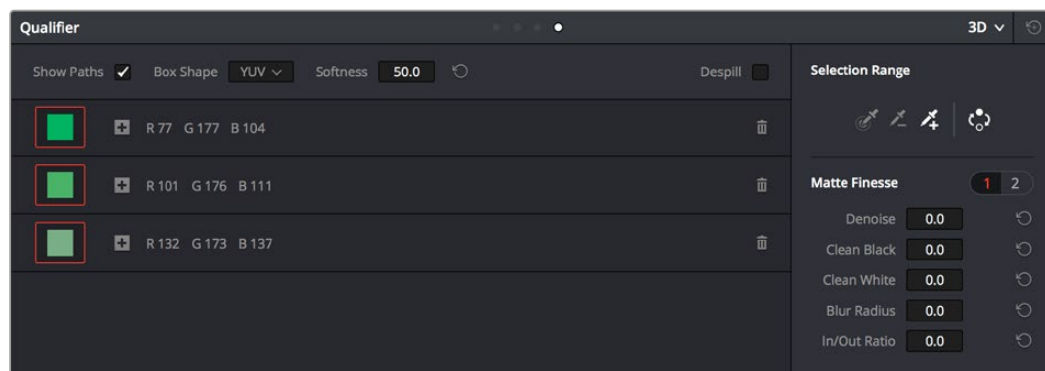
- 2 You'll want to use Node 1 to color correct the image to optimize it for keying. In this example, the foreground plate is log-encoded, so a LUT is added to Node 1 to normalize it.
- 3 Add a Serial node after Node 1 and connect its key output to the Alpha Output. This is a fast way to build this composite if you're confident that the corrections you've made in Node 1 won't adversely affect the key (or if you know they'll improve it by expanding the contrast of a flat-contrast source clip).

Alternately, you could create a second processing branch by adding a disconnected node (Node 2), connecting the Source node to its RGB input, and then connecting its key output to the Alpha Output you just added. Creating a second image processing branch from which to pull your key lets you key directly from the source (assuming the source is fit to key without adjustment), avoiding any problems that grading the first node could introduce to the key.



Node 1 grades the clip, expanding contrast to improve both the image, and its ability to be keyed. Node 2 uses Qualifier controls to pull the key.

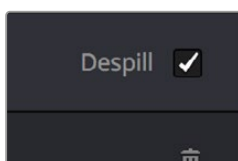
- 4 Use the Qualifier controls of Node 2 to key out the green background, and then turn on the Invert checkbox to create the proper composite. In this example, we're using the 3D Qualifier mode to pull a high-quality key.



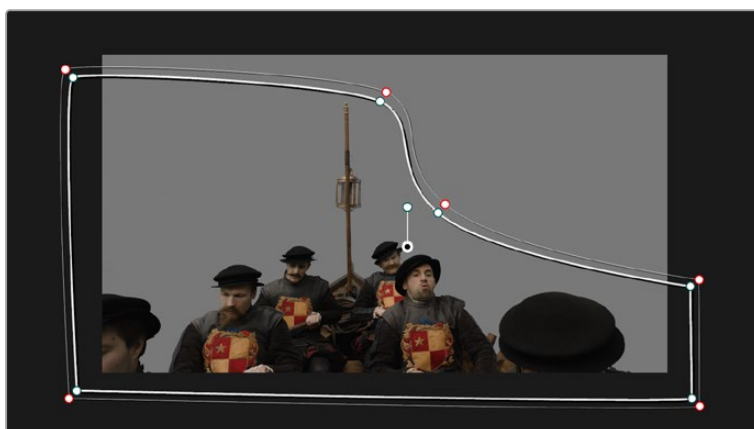


An HSL Qualification to key the green feeding the Alpha Output to create transparency

- 5 If there's green spill in the composite (or blue in cases where you're keying a bluescreen) you can turn on the Despill checkbox to eliminate it.

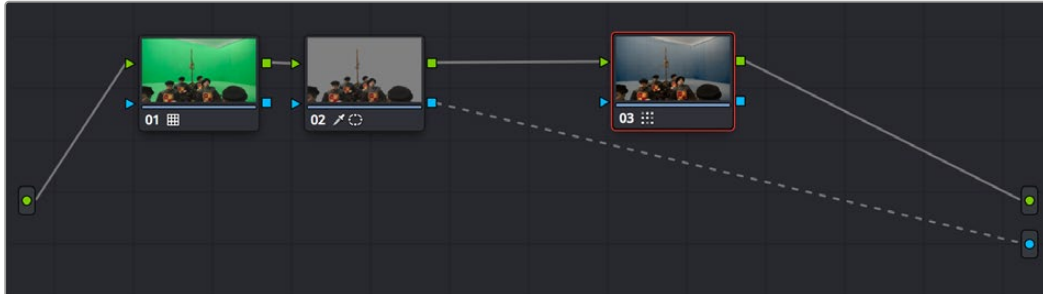


- 6 You can also use a Power Window to garbage matte out any elements you don't want intruding into the shot, using the Tracker palette if necessary to make the window follow the motion of the foreground subject.



A PowerCurve used to garbage matte out unkeyable lighting fixtures

- 7 If necessary, you can add another node to the output of Node 2 to make whatever corrections are necessary to the background plate clip to make the composite blend more seamlessly. For example, with this additional node selected, you could right-click the background clip and choose “Shot Match to This Clip” to use the automatic shot matching in DaVinci Resolve to adjust the foreground to match the background.



Adding more color adjustments to a node added after the qualifier

With all this set up, you end up with a nice green screen composite that’s visible in both the Color and Edit pages.



The result, with a window garbage matte cropping out the equipment, ready for further refinement

Using a Matte to Create Transparency

If you've been provided with a separate Matte clip for defining clip transparency, you can use that Matte clip within the node tree of a superimposed clip to create a composite using the Alpha Output.

To create a composite using an external matte:

- 1 To associate an external matte with a clip, open the Media page, select the clip you want to add a matte to in the Media Pool, navigate to the matte file using the Library browser, and then right-click the matte file and choose Add as Matte.

For more information on importing mattes, see *Chapter 146, "Combining Keys and Using Mattes."*

- 2 Open the Color page and select the foreground clip in the Thumbnail timeline, then right-click Node 1 in the Node Editor and choose the matte you added from the Add Matte submenu.

An EXT MATTE node appears underneath Node 1, with the first of its four key outputs connected to Node 1's key input. (Working with external mattes is covered in more detail in a prior section of this chapter.)

- 3 Now, right-click anywhere within the node graph (other than on a node) and choose Add Alpha Output from the contextual menu.

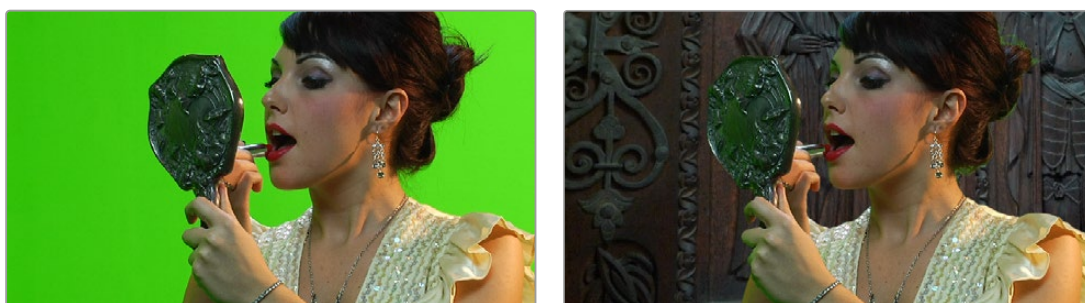
An Alpha output appears underneath the Node Tree output.

- 4 Drag a connection line from the second key output of the EXT MATTE node to the Alpha output.



Using an external matte in the grade and to the Alpha output

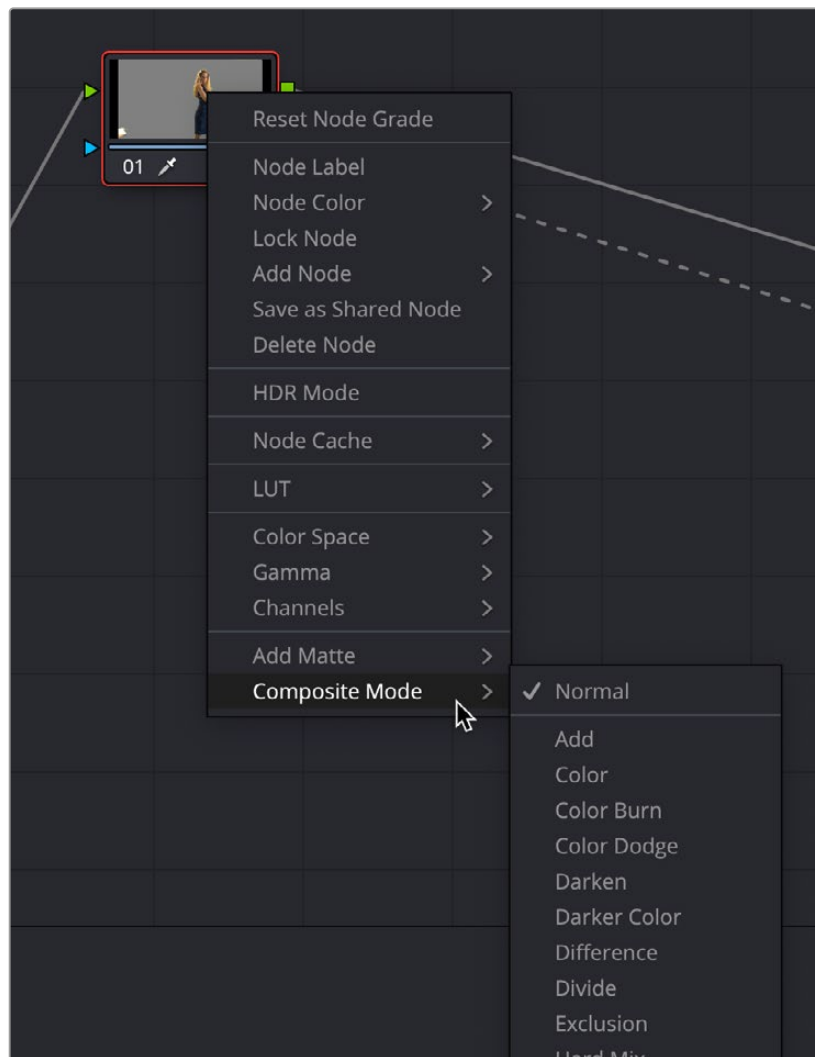
You should now see a successful composite, with the external matte creating a region of transparency in the foreground clip through which the background clip shows through.



Before and after the final composite, created by connecting an EXT MATTE node to the Alpha output in the node graph

Setting the Composite Mode in a Corrector Node

If you are compositing in the Color page using the Alpha output, you can set a specific composite mode for any corrector node. Right-click the node and select Composite Mode from the contextual menu. You can then change the type of composite for the node to any of DaVinci Resolve's built-in composite modes.



Choosing a Composite mode for the Corrector node

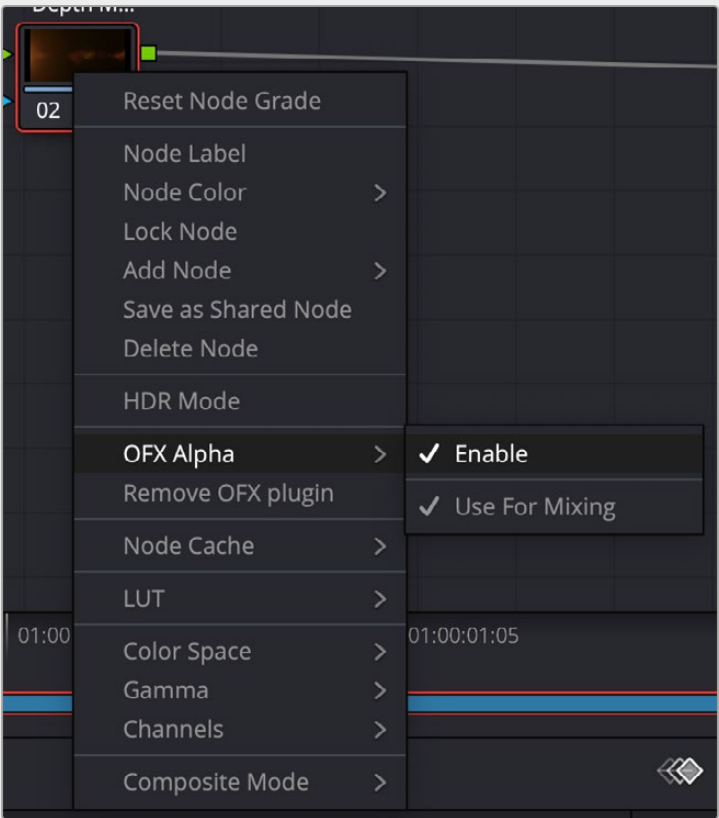
Using OFX Alpha Channels (Legacy)

Now that all effects use standard color corrector nodes, the OFX Alpha is always available without having to manually set a menu option. There is no need for the user to do anything else to access an OFX Alpha moving forward from DaVinci Resolve 18.5.

However, for legacy project use there is still an OFX Alpha menu with two options.

Enable: On by default. When this is unchecked, even if the effect generates an Alpha, the Alpha is ignored. Why one would go to the bother of generating an effect with an Alpha channel, only to discard it completely is a matter of conjecture, but the option is there if you're doing something exceedingly strange and complicated. Really just leave this on.

Use For Mixing: On by default. When unchecked, no mixing of the Alpha happens on the node at all. Power Windows and Keys stop working, and the node behaves like a legacy FX node. Any Alpha generated is still output, it just doesn't limit where the effect acts.



The Use OFX Alpha option has been replaced with the OFX Alpha menu. You most likely want to keep both Enable and Use for Mixing checked unless you have a really good reason.